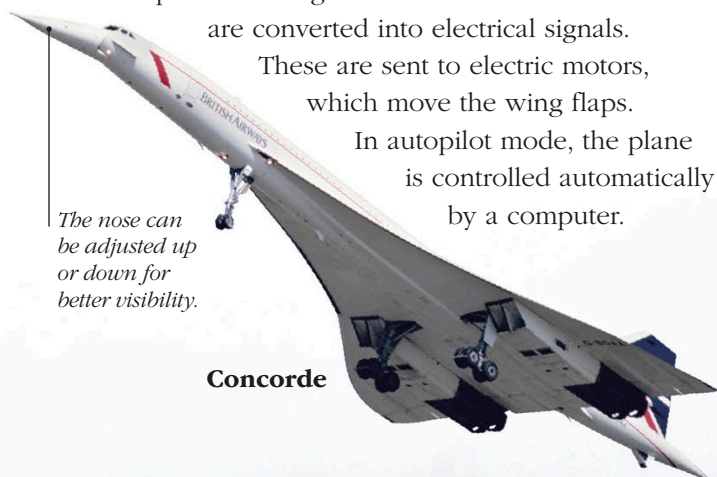


## FLY BY WIRE

In older planes, the wing flaps that control a plane's motion were moved mechanically by rods and levers. The supersonic (faster-than-sound) jet Concorde was one of the first "fly-by-wire" plane. In these, the pilot operates the flight controls and his movements are converted into electrical signals.

These are sent to electric motors, which move the wing flaps.

In autopilot mode, the plane is controlled automatically by a computer.



## GLASS COCKPIT

In the late 1960s, analog gauges gave way to electric displays. The display in modern airplanes is sometimes called a "glass cockpit" because it is full of screens that show data readouts, computer updates, and the plane's flight path.



## ◀ BOEING 314

Air travel was a luxury for the rich and the Boeing 314 was one of the most luxurious aircraft from 1938–1946. It could carry 74 passengers in seats or 40 in sleeping compartments. It also had a lounge, dressing rooms, and a dining room on board.

## FORCES OF FLIGHT

When a plane moves, its wings slice through the air at an angle, forcing the air downward and producing a force called "lift." This balances the plane's weight so it stays airborne. To keep the plane moving, the engines produce a force called "thrust," which balances the "drag" of the air.

